

1. Features

pre-exam record:
grades 10, 11 and
12 (term 1), school,
province, area
(F_{dt-cn} , 160 cols)

2. Center $\hat{y}$

XGBoost, squared
loss; bag of $B^* = 20$
(or one tuning
by RMSE)

3. OOF pairs

5-fold out-of-fold
($\hat{y}_{oof}, y$) on
training set; refit
on all training rows

4. Decision layer

for each K : fit g_K
on ($\hat{y}_{oof}, y$) with
weights $w_K(y)$
(R2 isotonic
or R1 bins)

5. Outputs

$g_K(\hat{y})$: decision
score under K ;
 $P(Y \geq c \mid x)$
for a fixed
threshold c

tail-weighted HPO fitness and
weighted training (R8) act here

output stretch (R5) and the other
layer estimators (R3, R4, R7) act here